

S5.2 Radian Measure and Angles on the Cartesian Plane

October 25, 2023 8:32 PM

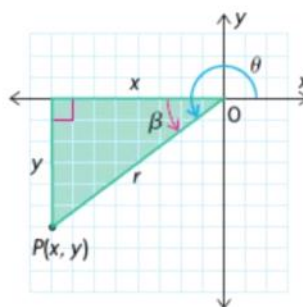


S5.2 Radian Measure and Angles on the Cartesian Plane

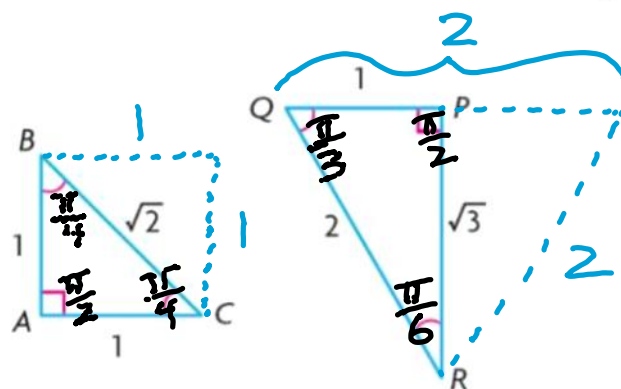
(5.2) Radian Measure and Angles on the Cartesian Plane

Recall:

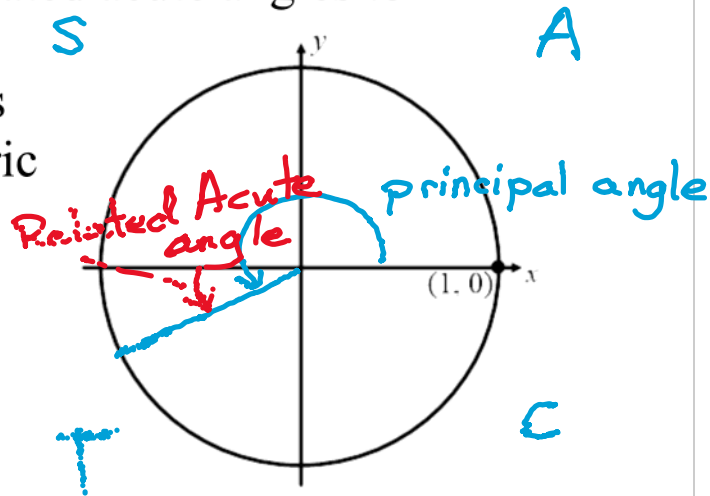
$$\begin{aligned} \sin \theta &= \frac{y}{r} & \csc \theta &= \frac{1}{\sin \theta} & \csc \theta &= \frac{r}{y} \\ \cos \theta &= \frac{x}{r} & \sec \theta &= \frac{1}{\cos \theta} & \sec \theta &= \frac{r}{x} \\ \tan \theta &= \frac{y}{x} & \cot \theta &= \frac{1}{\tan \theta} & \cot \theta &= \frac{x}{y} \\ \tan \theta &= \frac{\sin \theta}{\cos \theta} & \cot \theta &= \frac{\cos \theta}{\sin \theta} \end{aligned}$$



and

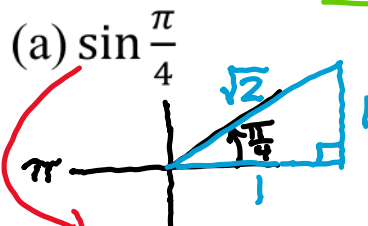


We use the Unit Circle, Special Triangles, the CAST Rule, principal angles, and related acute angles to understand different ratios of trigonometric values:



Ex. 1 Determine the exact value of each:

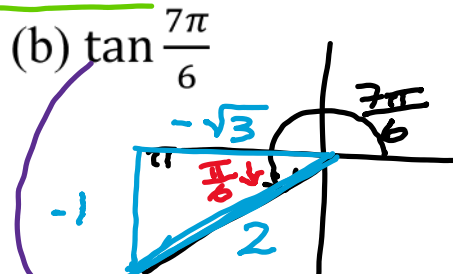
(a) $\sin \frac{\pi}{4}$



$$= \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{2}}$$

$$= \frac{\sqrt{2}}{2}$$

(b) $\tan \frac{7\pi}{6}$



$$= \frac{-1}{-\sqrt{3}} \cdot \frac{\sqrt{3}}{\sqrt{3}}$$

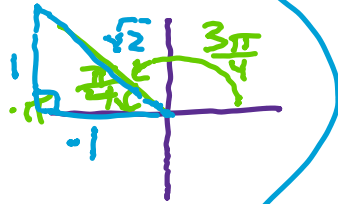
$$= \frac{\sqrt{3}}{3}$$

$$\pi + \frac{\pi}{6}$$

$$= \frac{6\pi}{6} + \frac{\pi}{6}$$

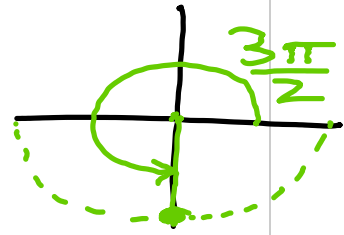
$$= \frac{7\pi}{6}$$

$$(c) \csc \frac{3\pi}{4}$$



$$\begin{aligned}
 &= \frac{1}{\sin \frac{3\pi}{4}} \\
 &= \frac{1}{\frac{1}{\sqrt{2}}} \\
 &= \frac{\sqrt{2}}{1} \\
 &= \sqrt{2}
 \end{aligned}$$

$$(d) \cot \frac{3\pi}{2}$$



$$\begin{aligned}
 r &= 1 \\
 x &= 0 \\
 y &= -1
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{0}{-1} \\
 &= 0
 \end{aligned}$$

Ex. 2 If $\cos \theta = -\frac{5}{13}$ where

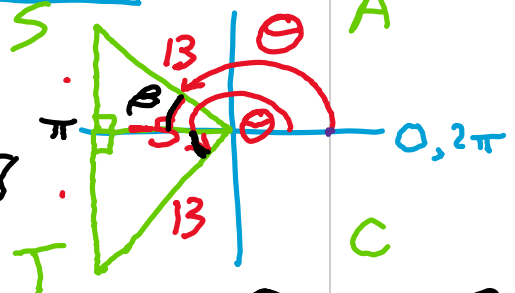
$0 \leq \theta \leq 2\pi$, evaluate θ to the nearest hundredth.

$$\theta = \cos^{-1}\left(-\frac{5}{13}\right) \approx 1.97$$

A

$$\cos \beta = \frac{5}{13}$$

$$\beta \approx 1.18$$



$$\theta = \pi - \beta$$

$$\theta \approx 1.97$$

$$\theta = \pi + \beta$$

$$\approx 4.32$$

C

Ex. 3 State $\cos\left(\frac{4\pi}{3}\right)$ as an equivalent expression in terms of the related acute angle.

$$\cos\left(\frac{4\pi}{3}\right) = -\cos\left(\frac{\pi}{3}\right)^S$$

